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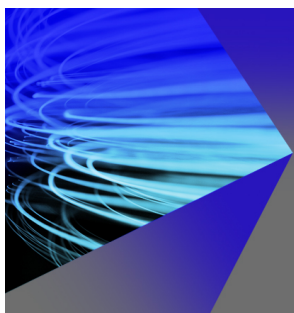
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Hysteretic interaction of conical liquid sheets from coaxial atomizers: Influence on the spray characteristics

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In the context of coaxial swirl injectors used in liquid bipropellant rockets, a detailed experimental study has been made on the mutual interaction between thin coaxial, conical liquid sheets. As the flow rates are increased, the sheets move close to each other and merge at a definite flow condition. The presence of the inner jet influences the behavior of the outer jet, even before merging. The merging and separation processes exhibit hysteresis. This leads to drastic variations in spray characteristics depending on the approach to the hysteresis regime. Many of these features as well as those on the movement of the contact point can be traced to changes occurring in the static pressure in the cavity between the two sheets. © 1998 American Institute of Physics.

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I. INTRODUCTION

Liquid-liquid coaxial swirl atomizers are widely used in liquid propellant rocket engines to take advantage of their higher mixing efficiency within a short length of the combustion chamber. The atomizer is comprised of an inner orifice and an outer orifice to release the oxidizer and fuel sprays, respectively, into the combustion chamber. In this atomizer configuration, the jets are released in the form of thin coaxial conical liquid sheets. Merging of the liquid conical sheets and the subsequent mixing of the two jets are essential for the atomization process. In actual working conditions of an engine, the liquid sheets merge and mix with each other very close to the exit of the orifices.

Under low flow rate conditions, the two liquid sheets are separated from each other, as seen in Fig. 1(a). A gradual increase in the liquid flow rate through the orifices induces the liquid sheets to interact and merge with each other at some particular flow conditions and result in a single conical jet, as shown in Fig. 1(b). A further increase in the liquid flow rate retains the sheets continuously in the merged form. Conversely, a gradual decrease in the flow rate from the merged flow rate condition leads to the separation of liquid sheets at a definite low flow condition. This phenomenon of merging and separation of the liquid sheets assumes importance while throttling is effected in an engine. Detailed information on such an interaction of the liquid sheets is very important for any improvement in the throttling performance characteristics of an atomizer. There appears to be very little work on a liquid-liquid jet interaction that involves several interesting phenomena. In the present experimental work we investigate the physical process underlying the merging and separation behavior of the liquid conical sheets formed from a coaxial swirl atomizer, while variation in flow conditions are effected. The role of geometrical configuration of the atomizer in the phenomenon is also delineated.

A large number of theoretical and experimental studies have been made on the jet interaction, with many variations

like the interaction of a jet with nearby surfaces, the interaction of dual and multiple jets, the interaction of equal and unequal buoyant plumes, the turbulent interaction of concentric jets in the absence and presence of the swirl effect, the interaction of an annular air jet on a central liquid jet, etc.¹⁻¹⁷ Many of these investigations deal mainly with the flow features arising out of the jet interaction with either a solid surface or an adjacent jet through the entrainment process.¹⁻¹⁰ In the jet-jet interaction dealing with gaseous jets,⁵⁻¹⁰ the jets merge at a smaller distance from the exit plane by bending under a mutual influence and exhibit the behavior of a single jet after the merging point. In this case the flow field can be classified into three distinct regions: the converging region, the merging region, and the combined region. A subatmospheric zone is established between the jets in the converging region, owing to the mutual entrainment process. It seems possible to control the merging point through the controlled supply of fluid into this subatmospheric zone to compensate for the amount of fluid entrained by the jet, which could prevent the drop in static pressure inside the zone.⁹ It is also shown that this subatmospheric region is influenced strongly by the flow conditions of the individual jets.^{8,10} Ko and Kwan¹¹ have obtained more information about the converging and merging regions through turbulence measurements. Dahm *et al.*¹³ and Rehab *et al.*¹⁴ have provided further detailed information about the near-field interaction behavior of different mixing layers by means of flow visualization through laser-induced fluorescence technique.

Recently more attention has been paid to study the influence of jet interaction on the spray characteristics coaxial atomizers.¹⁵⁻¹⁷ The influence of the liquid-gas interaction on the disintegration behavior of the central liquid jet has been examined by Mayer and Krulle¹⁵ experimentally and compared with numerical simulation. It is reported that the liquid jet turbulence is responsible for the initiation of the wave pattern, whereas the flowing ambient gas is responsible for the eventual atomization of the liquid jet. However, the in-

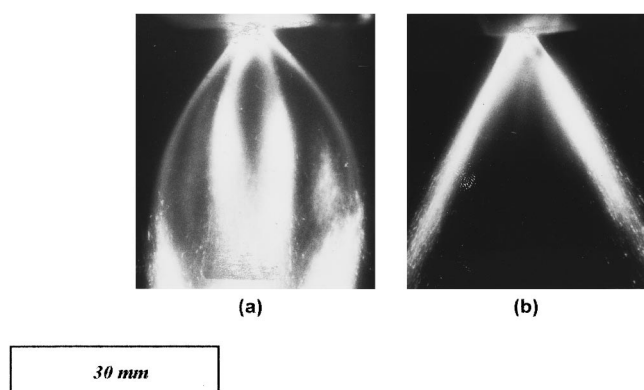


FIG. 1. Images of the conical jets during transition, $Re_o=9724$ (atomizer configuration index—42). (a) $Re_i=11\,349$, and (b) $Re_i=11\,516$.

roduction of the swirling effect to the outer gas jet has been shown to alter the behavior of the nonswirling liquid jet by causing rapid disintegration and radial expansion through the establishment of a strong recirculatory motion.¹⁶ In the case of a liquid sheet, the disintegration behavior is drastically altered by the interaction of unswirled airflow by causing major disturbances over the liquid sheet.¹⁷ A common observation is that the liquid jet characteristics are strongly influenced by the external air jet conditions in the vicinity of the orifices.

The fluid dynamic interaction of liquid jets from a liquid–liquid coaxial atomizer and its influence on the spray characteristics could be quite different from those observed in liquid–gas coaxial atomizers. Although a few studies^{18–22} on the liquid–liquid jet interaction have been reported, none has described the several features such as hysteresis, drastic variations in jet dimensions, and consequent drop size distributions. The underlying physical mechanism remains unexplained. Amagai and Arai¹⁸ studied the disintegration process of liquid–liquid nonswirling coaxial jets through experiments and concluded that the breakup length of the combined jet is influenced by the flow conditions of both the inner and the outer liquid jet. Seol *et al.*²¹ studied the interaction of liquid conical sheets formed from dual-orifice fuel injectors, with the outer orifice projecting beyond the exit of the inner orifice. Their flow visualization study does show mutual influence, but the study is restricted to low-pressure drop conditions at which the jets are not diverging. Recently, the present authors²² studied the fluid dynamic interaction of liquid sheets emanating from a liquid–liquid coaxial swirl atomizer and confirmed that the two liquid conical jets influence each other's spray characteristics, like the spray cone angle and disintegration distance, even before they merge to become a single conical spray. In the present work we look at the phenomenon in greater detail and examine the mechanisms underlying this interaction and describe qualitatively and quantitatively the spray characteristics before and after the merging process.

II. EXPERIMENTAL PROCEDURE

A. Coaxial atomizer details

Experiments have been conducted by generating coaxial conical liquid sheets using specifically designed coaxial swirl

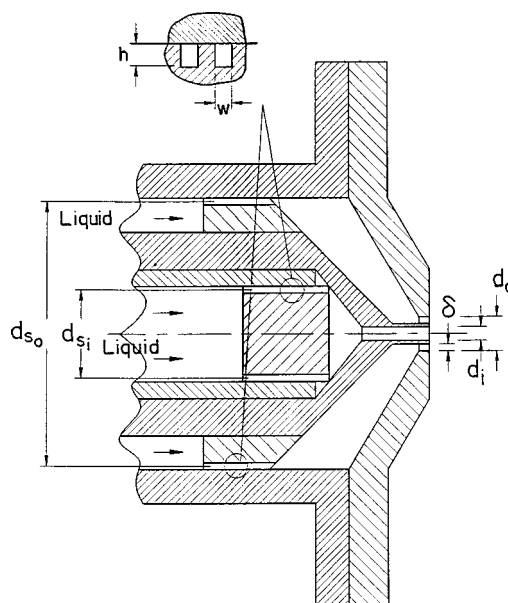


FIG. 2. Schematic of the coaxial swirl atomizer with a swirler element.

atomizers with water as the experimental fluid. The schematic of the coaxial swirl atomizer with some principal dimensions is shown in Fig. 2. The contraction ratio (the ratio of swirl chamber diameter to the orifice inlet diameter) for the inner nozzle geometry is 8:1 and those of outer nozzle geometry are 10.6:1 and 7.7:1 for the two different orifice diameters, $d_o=2.7$ and 3.7 mm, respectively. The orifice diameter of the inner atomizer, d_i , is 1.0 mm and the outer diameter of the inner atomizer orifice is 1.95 mm. The need for providing more length to the inner orifice arises from the practical requirement of establishing undisturbed outer swirling flow between the two convergent walls (see Fig. 2). In the case of pressure swirl atomizers, the flowing liquid stream always remains in contact with the outer wall of the orifice due to the centrifugal effect. Hence, any marginal variation in orifice length is unlikely to influence the jet characteristics significantly, except for the marginal rise in frictional losses. This would reflect in the discharge coefficient without much altering the sheet formation outside. So any minor alteration of these geometrical parameters is not expected to modify the observed phenomena during the liquid–liquid sheet interaction reported in the following sections.

Extensive care has been exercised to ensure that the orifices are coaxial and coplanar. Any misalignment between the orifices can affect the symmetry and swirling motion of the conical jets and, hence, the subsequent interaction between them. A provision has been made in the atomizer to place both inner and outer orifices coaxially through an adjustment of the outer orifice plate. Before each experimental run, the concentricity of the orifices is confirmed by examining the exit plane using a microscope having a magnification factor $10\times$. The liquid flow rate is controlled by fine needle valves and monitored with standard analog pressure gauges having a resolution of 0.5 psi. A low-pressure drop filter has been introduced in the liquid flow path to arrest any contaminant that may clog the atomizer flow passages.

The swirling motion is generated by passing the liquids

TABLE I. Different geometrical configurations of a coaxial swirl atomizer.

Atomizer configuration index	Inner orifice					Outer orifice				
	d_i mm	w_i mm	h_i mm	S_i	$C_{d,i}$	d_o mm	w_o mm	h_o mm	S_o	$C_{d,o}$
11	1.0	1.0	1.0	2.8	0.39	2.7	1.0	1.0	15.1	0.26
12	1.0	1.0	0.6	4.8	0.30	2.7	1.0	1.0	15.1	0.26
21	1.0	1.0	1.0	2.8	0.39	2.7	1.0	0.6	25.6	0.16
22	1.0	1.0	0.6	4.8	0.30	2.7	1.0	0.6	25.6	0.16
31	1.0	1.0	1.0	2.8	0.39	3.7	1.0	1.0	36.7	0.10
32	1.0	1.0	0.6	4.8	0.30	3.7	1.0	1.0	36.7	0.10
41	1.0	1.0	1.0	2.8	0.39	3.7	1.0	0.6	62.1	0.06
42	1.0	1.0	0.6	4.8	0.30	3.7	1.0	0.6	62.1	0.06

through rectangular helical passages, as shown in Fig. 2. Double start helical passages have been used in the swirlers to improve the uniformity of the conical jet. The swirl intensity given by swirl number, S (see Ref. 23 for details) takes the form

$$S_i = \frac{\pi d_i d_{s,i}}{4 n_i w_i h_i},$$

for the inner jet and

$$S_o = \frac{\pi \delta d_{s,o}}{n_o w_o h_o},$$

for the outer jet, where d and d_s represent the orifice exit diameter and the mean diameter of the swirler, respectively, w and h the width and the depth of the rectangular helical passage, δ the outer orifice gap, and n the number of starts in the swirler. Here the subscripts “ i ” and “ o ” refer to the inner and outer jet, respectively. Various combinations of orifices, inner, and outer swirlers used are presented in Table I. Note that changes in S occur through the swirler passage depths and the outer orifice diameter, d_o .

The liquid flow rate, $\dot{m}$ through the orifices is measured by collecting the liquid for a known duration. The discharge coefficient, C_d , of the orifices are determined from the expression

$$C_d = \frac{\dot{m}}{A_e \sqrt{2 \Delta P \rho_l}},$$

where A_e is the exit orifice area, ΔP the pressure drop across the atomizer, and ρ_l the liquid density. For the present atomizer geometry, the mechanism of liquid sheet formation is similar to the simplex pressure swirl atomizer.²⁴ The general expression for the thickness of the liquid sheet at the orifice exit, t_f , based on atomizer geometry and flow parameters, is given by²⁴

$$t_f = 3.66 \left(\frac{d \dot{m} \mu}{\rho_l \Delta P} \right)^{0.25},$$

where μ is the liquid dynamic viscosity. The liquid flowing area, A , is estimated from the calculated t_f as $\pi(d - t_f)t_f$. The Reynolds number, Re , for the test condition is calculated from the standard expression $Re = \rho_l V L / \mu$, where the length scale $L = d_i$ for the inner jet and $L = 2\delta$ for the outer jet

based on the hydraulic diameter. The term V is the mean axial velocity at the orifice exit derived from $\dot{m}$ and A . The Weber number, We , is calculated as $We = \rho_a V^2 t_f / \sigma$, where ρ_a and σ are the density of air and liquid surface tension, respectively. The mean fluid properties used for the calculations are $\rho_l = 1000 \text{ kg/m}^3$, $\rho_a = 1.225 \text{ kg/m}^3$, $\mu = 0.00085 \text{ kg/m s}$, and $\sigma = 0.0725 \text{ N/m}$. In the range of test conditions, C_d is found to be nearly constant for all combinations. These mean C_d values are also given in Table I.

B. Visualization

The still photographic technique has been used to visualize the structure of the spray. A 35 mm still camera is used to take color and black and white photographs with a shutter speed of 1/15 s using 400 ASA films. A plane of symmetry of the jets is illuminated using a thin collimated sheet of light of thickness approximately 1 mm generated from a source placed at right angles to the line of sight of the camera. A sodium-based fluorescent dye was added to the experimental liquid to improve the contrast in the pictures. These photographs clearly delineate the contours of both the conical jets, disintegration and contact regions, as well as the merging and separation conditions of the liquid sheets, which provide the primary data and observations for the present study. A backlighting technique has been used for low shutter speed (1/1000 s) to obtain high-contrast pictures. Additional information on the jet interaction is obtained through 16 mm cine photography (64 frames/s) using a backlighting technique and also through visual observation of the sheets by illuminating the spray with a strobe light. Quantitative data from these photographs are obtained using a 16 mm picture analyzer and a microfilm reader with a magnification factor of 17.5 $\times$.

C. Drop size measurement

Drop size distributions have been measured using the Malvern mastersizer X instrument, which works on the Fraunhofer forward diffraction principle. The size distribution is obtained from the scattered light intensity. This instrument provides the line of sight size distribution, mean drop size (SMD, Sauter mean diameter) and liquid volume concentration. The principle of operation of the instrument is described in Ref. 25.

III. RESULTS AND DISCUSSION

A. Behavior of individual jets

A number of preliminary experiments have been conducted to observe the behavior of individual jets. Basic spray parameters like the spray cone angle, disintegration distance, drop size distribution, etc., are obtained for all atomizer configurations. Photographs of the sheet disintegration obtained with short (1/1000 s) as well as long exposure times (1/15 s) and from various angles reveal that the disintegration process is uniform and repeatable. In the lower range of $\dot{m}$, a converged or tulip-shaped jet is formed, owing to the domination of surface tension and the pressure differential over inertia. A fluctuation in the length of the tulip is observed

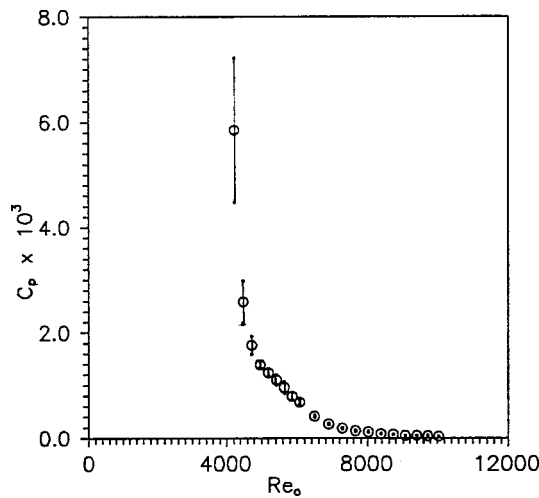


FIG. 3. Variation of pressure coefficient, C_p , across the outer liquid sheet against Re_o (atomizer configuration index—42).

under these flow conditions, which may be attributed primarily to the pressure fluctuations inside the converged jet. An attempt has been made to measure the pressure variation inside the outer jet. A low range digital micromanometer (0–20 PI) of accuracy 0.01 PI has been used to extract the pressure data. Here, the inner orifice has been used as the pressure tapping, and therefore the measurement location corresponds to the exit of the outer orifice. The minimum and maximum values of pressure difference levels have been recorded visually from the micromanometer for different $\dot{m}$. The variation of the mean value of the pressure coefficient, $C_p = (P_e - P_i) / \frac{1}{2} \rho_l V^2$ (where P_e and P_i are the pressure outside and inside the liquid sheet, respectively) against the outer jet Reynolds number, Re_o , is shown in Fig. 3. In the figure the maximum and minimum limits of C_p variation are indicated by bands, and it is observed that the pressure fluctuation is more at a lower range of Re_o . The observations made above point toward the close relation between pressure variations and the sizes of the cavities formed within liquid sheets.

The typical variations of spray cone angle and SMD of the individual conical jets are plotted in Figs. 4(a) and 4(b)

against Re . The spray cone angle of the jet is determined from the spray width measurement at 10 mm distance from the exit orifice plane, SW_{10} , as suggested in Ref. 26. The theoretically estimated liquid film thickness variation based on the equation mentioned above is also shown in Fig. 4(a) as a function of Re . The qualitative trends of the variation of spray parameters conform to the trends of pressure swirl atomizers described in Ref. 26. Note that the individual spray parameters of each jet have been obtained with the coaxial and coplanar exit orifice geometry, but without the presence of the other jet.

B. Influence of the inner jet on the spray contour of the outer jet

The changes caused in the outer conical sheet by varying the inner jet Reynolds number, Re_i , is shown in Fig. 5 for a particular Re_o . Figure 5(a) depicts the configuration of the outer jet in the absence of an inner jet. The presence of an inner jet in the core of the outer jet alters its contour by bending the liquid sheet toward the spray axis, as seen in Fig. 5(b). With an increase in Re_i , the outer sheet contracts farther and moves closer to the expanding inner liquid sheet as in Fig. 5(c), which is close to the state of merging. A further increase in Re_i makes the two conical sheets merge with each other and then disintegrate into drops, as was observed in Fig. 1(b). The presence of an inner jet in the core of the outer jet entrains the air between the liquid sheets and thereby reduces the static pressure. However, this gives rise to the recirculation of air trapped between the sheets, which is augmented by the swirling effect. With increasing Re_i , the possibility of ambient air entering this region diminishes, which leads to an increase in $(P_e - P_i)$, due to a drop in static pressure between the liquid sheets, and this is accompanied by contraction of the jet.

The variation of the outer jet spray width arising from the above-described mechanism is plotted in Figs. 6(a) and 6(b) at two discrete axial stations, i.e., a 10 and 30 mm distance from the exit orifice plane for different Re_o . The data have been nondimensionalized with respect to outer orifice gap, δ . Theoretical results of Ramos²⁷ show that at any downstream location, the width of the liquid sheet reduces

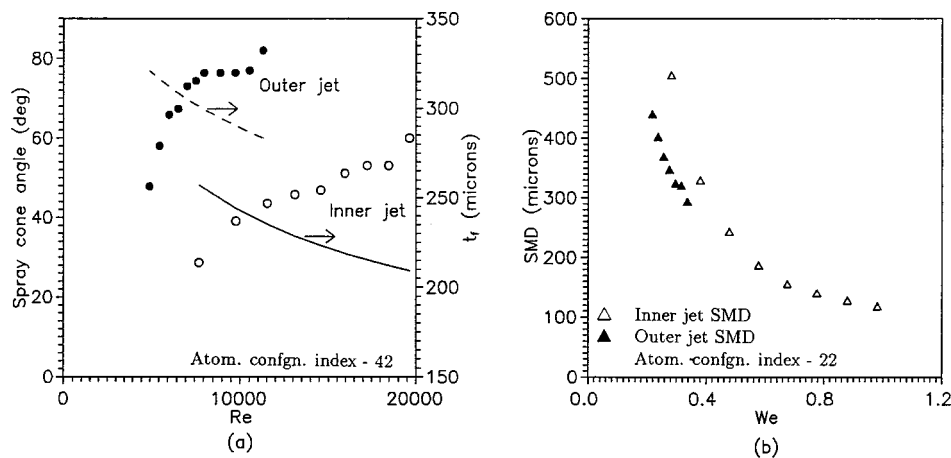


FIG. 4. Typical variation of spray parameters of individual jets.

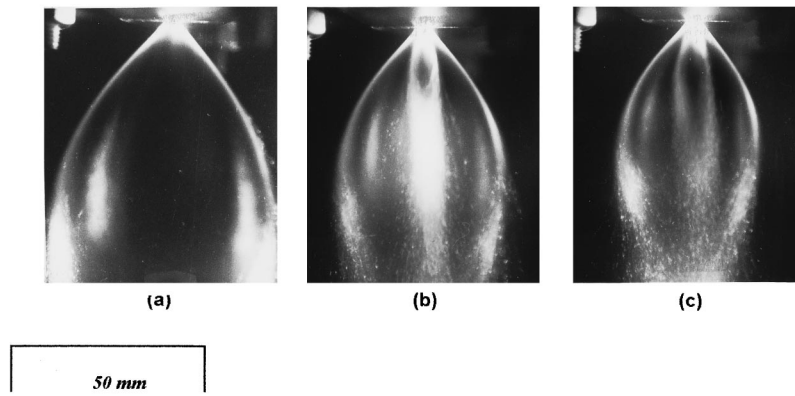


FIG. 5. Influence of the inner jet on the spray contour of the outer jet with $Re_o = 10\,515$ (atomizer configuration index—41). (a) $Re_i = 0$, (b) $Re_i = 10\,790$, and (c) $Re_i = 12\,227$.

with the increase in $(P_e - P_i)$, and vice versa. In the present study, P_i , for the outer sheet, which is equal to P_{int} , the intermediate pressure between the liquid sheets decreases with increasing Re_i due to the entrainment process, and hence reduces the spray width, as seen in Figs. 6(a) and 6(b). In contrast, the spray width for the inner liquid sheet may be expected to increase relative to the value in the absence of the outer jet, owing to the reduction of $P_e (=P_{int})$, which, however, cannot be easily measured. As a common practice, the spray cone angle of a conical sheet is estimated from the spray width measurement at a distance of 10 mm from the exit orifice plane.²⁶ It is observed from Figs. 6(a) and 6(b) that at a 10 mm distance, the influence of the inner jet on the spray width is not discernable, though the spray cone is visibly altered. But the measurement of the spray width at a farther station, at 30 mm, shows the influence of the inner jet clearly.

The influence of the inner jet on the shape of the outer converged jet (tulip bulb stage) is depicted in Fig. 7 evidently for a low Re_o . The presence of the inner jet at a lower Re_i initially elongates the outer tulip as in Fig. 7(b) relative to the length, in the absence of the inner jet seen in Fig. 7(a). This may be attributed to the slight increase in the pressure inside the tulip because the activated inner jet effectively reduces the free volume. However, at higher Re_i , even as the inner jet forms its own tulip, there is a gradual reduction in

both the length and the width of the outer tulip, as can be seen in Figs. 7(c) and 7(d). The last one is very close to the merging condition. This behavior is attributed to the decrease in static pressure between the two sheets, as explained previously.

C. Influence of the inner jet on the disintegration distance of the outer jet

As discussed before, the static pressure drop in the core of the outer jet induces the liquid sheet to bend inward and also to disintegrate at a shorter axial distance. A set of experiments has been conducted specifically to study the influence of the inner jet on the disintegration process of the outer liquid sheet. The disintegration location of the outer liquid sheet has been determined based on the color contrast in the photographs taken by adding sodium fluorescence to the experimental liquid. For example, a change in shade in Fig. 5(a) indicates the disintegration location of the outer liquid sheet. The variation of the nondimensionalized breakup length for different outer jet conditions is plotted against Re_i in Figs. 8(a) and 8(b) for different atomizer configurations. It may be inferred from the plots that the outer jet breakup length is strongly influenced by the inner jet as the merging state is approached. Photographs obtained at a low shutter speed delineates this phenomenon clearly. A visual observa-

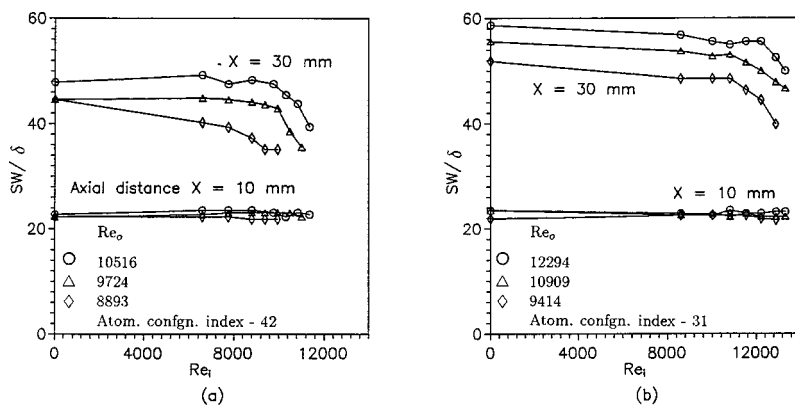


FIG. 6. The outer jet spray width variation with Re_i .

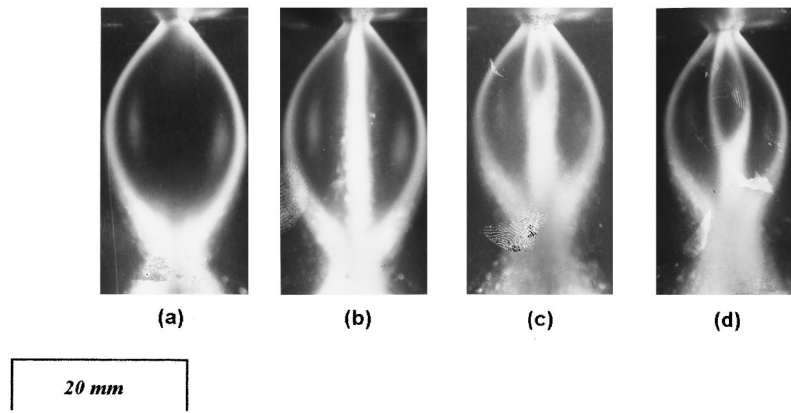


FIG. 7. Photographs showing the effect of the inner jet on the shape of the outer converged jet with $Re_o = 6591$ (atomizer configuration index—42). (a) $Re_i = 0$, (b) $Re_i = 3575$, (c) $Re_i = 6325$, and (d) $Re_i = 8172$.

tion of the spray reveals that the amplitude of the waviness or the perturbations on the liquid conical sheet increases at higher Re_i . This again is a consequence of the increasing static pressure difference. In the process, the sheet gets disrupted and breaks in a shorter distance.²⁸ Seol *et al.*²¹ have, however, mentioned that the influence of ejection of the inner jet in the core of the diverged outer jet was not discernable. This is perhaps because of the distinct orifice configuration of the injector used in their experimental study. This is discussed further in the following sections.

D. Merging and separation behavior of the liquid sheets

In the case of straight jets without a swirling effect,^{5–10} the merging of jets is inevitable at all flow conditions, with minor changes in the flow features. In contrast, the merging of conical liquid sheets occur at definite flow conditions of inner and outer jets. It is to be noted that, with changes in the flow conditions, the spray characteristics of the individual jets, like the spray cone angle, disintegration behavior of liquid sheets are significantly altered, as mentioned in the previous sections, which could also influence the merging behavior between them.

1. Transition process from the separated to the merged state

The merging process of liquid sheets is accomplished through a well-behaved rapid transition process from the separated to merged state, which occurs at a specific Re_i for a particular Re_o . This inner flow condition is referred to as the merging flow condition for that particular Re_o . This transition process is illustrated in Fig. 9 through cine photographs for a typical Re_o . Note that the flow conditions have been held unchanged during the transition process. The liquid sheets initially make contact at a point quite distant from the exit orifice plane, as observed in the second and third frame of Fig. 9. As soon as the contact is established, the liquid sheet contact point or the merging point moves very rapidly toward the exit orifice plane, as seen in the succeeding photographs of Fig. 9. At the end of the transition process, a final steady-state merging distance from the exit orifice plane, X_{mp} , is reached, as shown in the last frame of Fig. 9. A further analysis of cine films reveals a slight oscillation of the merging point after the transition process. This phenomenon is under investigation and therefore has not been discussed further in this paper.

Just as the contact between the sheets is established, the air between the liquid sheets is entrained out rapidly through

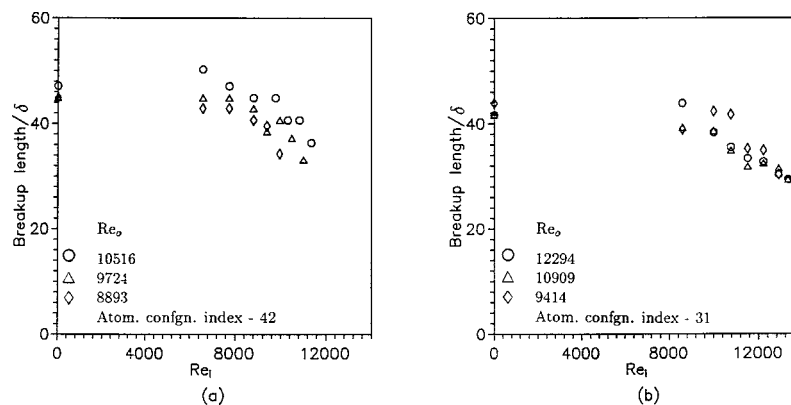


FIG. 8. Variation of the outer liquid sheet breakup length with Re_i before merging.

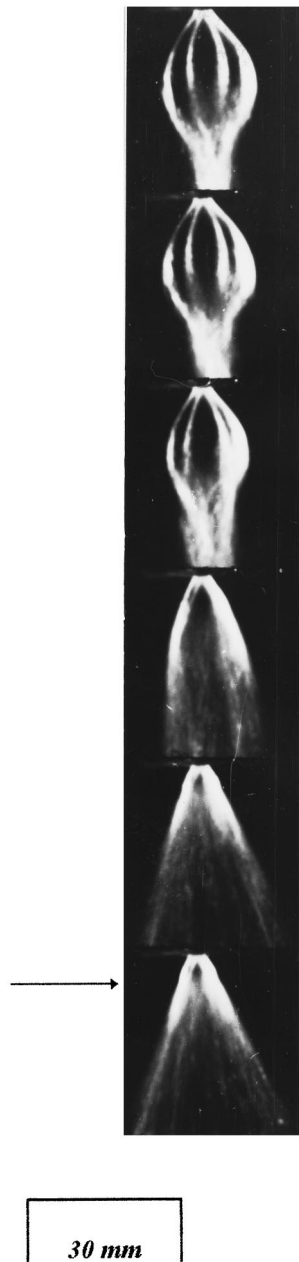


FIG. 9. Cine photographs showing the transition process from the separated to the merged flow state for a typical outer flow condition, $Re_o=7558$ and $Re_i=10\,127$ (atomizer configuration index—42).

the liquid sheet contact region. This entrainment process is primarily associated with the sudden increase in a local turbulent intensity level at the liquid sheet contact region arising out of intense mixing of two swirling jets. However, no quantitative measurements of this turbulent intensity have been obtained in the present investigation. An attempt has been made to justify the above qualitative argument through the reported literature. In jet–jet interaction studies, many investigators have observed a steep increase in the turbulent intensity level as the merging region of the jets is approached. The reported measurements clearly demonstrate the existence of a peak in turbulent intensity at the merging region.^{6–8} Further, it has been shown that the turbulent inten-

TABLE II. Number of frames covered during transition from separation to merging (atomizer configuration index—42 and film speed—64 frames/s).

Re_o	Re_i	No. of frames	Time taken (ms)
6077	8065	10	156
6591	8689	8	125
7083	9475	9	141
7558	10 035	6	94
8017	10 486	5	78
8893	11 009	5	78

sity level in the jets has a direct effect on the extent of entrainment.

So the entrainment process reduces the static pressure, P_{int} , in the cavity during the transition process. Since there is no path for the ambient gas to enter into the cavity, the merging point moves upstream, depending on the pressure gradient established at the liquid sheet contact region. It is also observed that the transition process is very much influenced by the flow conditions of the jets. Table II, which shows the number of frames and time taken by the transition process to attain the final steady-state X_{mp} , reveals that the movement of the contact point during this process is accelerated with the increase in Re_o . This may be attributed to the variation of the local turbulent intensity level at the liquid sheet contact region that is responsible for the rate of entrainment of air from the cavity.

2. Influence of flow conditions on the merging and separation process of liquid sheets

The Re_i at which the merging and separation of liquid sheets occur is plotted against the outer jet Re_o in Fig. 10 for one of the atomizer configurations. The increase in the outer jet spray cone angle with Re_o is responsible for the variation seen in Fig. 10. As a consequence, the Re_i values for both merging and separation increase with Re_o and approach constant values at higher Re_o . It can be inferred from Fig. 10

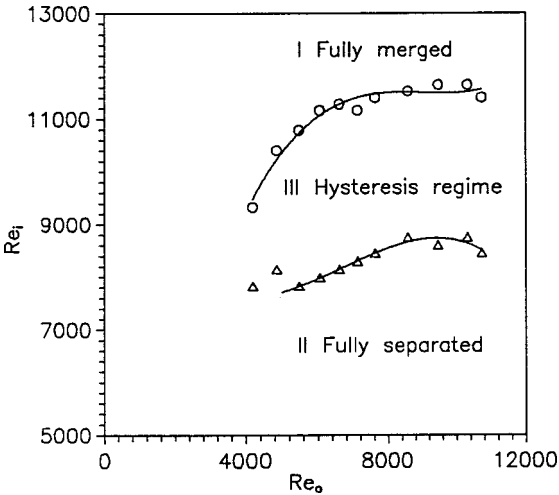


FIG. 10. Flow conditions for the merging and separation process for different outer jet flow conditions ($\circ$, merging condition; $\triangle$, separation condition; atomizer configuration index—11).

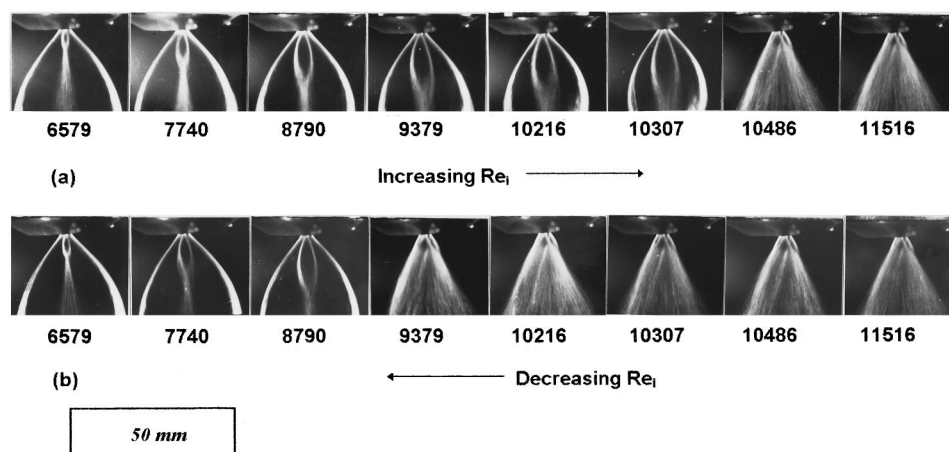


FIG. 11. The photograph sequence showing the hysteresis phenomenon during merging and separation of liquid sheets with $Re_o = 9314$ (atomizer configuration index—42).

that the merging and separation of liquid sheets occur at two distinct Re_i for a particular Re_o and thereby exhibit hysteresis.

Figure 11 presents a series of images to illustrate the hysteresis phenomenon qualitatively. These images correspond to a particular Re_o with different Re_i . The images presented in Fig. 11(a) are for increasing values of Re_i . The configuration of jets at the merged flow condition can be seen in the seventh frame in Fig. 11(a). With a further increase in Re_i , the jets retain the same configuration. In Fig. 11(b), the corresponding jet configurations are shown for the same flow conditions as in Fig. 11(a), but for a decreasing value of Re_i . The separation of liquid sheets during this process can be seen in the third frame of Fig. 11(b) that occurs at Re_i , which is different from that for merging. By comparing the photographs of Figs. 11(a) and 11(b), one can easily observe the existence of different jet configurations for the same Re_i depending on the mode of operation.

As the liquid sheets contact, the air between them is entrained out accompanied by a reduction in P_{int} , as explained previously. The process of separation now will have to encounter a lower suction pressure in the cavity compared to that at the time of contact. This calls for lower inner flow conditions (reduction in momentum leading to the convergence of the jet and thereby pulling the liquid sheet away from the outer liquid sheet) in comparison with the merging flow conditions. This results in hysteresis as observed in Figs. 10 and 11. Note that the hysteresis arises here due to the reduction in P_{int} during the transition process. Similar hysteresis phenomenon has been reported by Bourque and Newman¹ in connection with the reattachment of a jet with an adjacent inclined plate. In their work, at a specific inclination of the plate, the jet is separated or reattached depending on how the flow is started. However, the authors have not paid much attention to discussing the physical features of the phenomenon in detail, as the phenomenon is of no interest to their investigation.

Thus, in a liquid-liquid coaxial swirl atomizer, the jets could be in any of the three regimes as shown in Fig. 10. In regime I, which occurs at higher Re_i , the jets are always in a

merged state and form a single conical jet with the liquid sheet contact region close to the exit of the orifice. In contrast, the jets are always in the separated state in regime II at lower Re_i . In regime III, the jets are in either a merged or separated state depending on how the corresponding flow conditions are attained. If the inner jet flow rate is in a decreasing mode, then the jets will be in merged state, otherwise they will be separate.

3. Influence of Re_o on the final steady-state merging distance during the merging process

As discussed in the transition process of liquid sheets from the separated to the merged state, the process of merging establishes a contact region upstream of which the two liquid sheets maintain their identity. It is only on the downstream of the contact region that a single combined jet develops and eventually disintegrates. Figure 12 shows the typical variation of the final steady-state X_{mp} (see the last frame of Fig. 9) just above the merged flow condition against Re_o for different atomizer configurations. In this figure, each

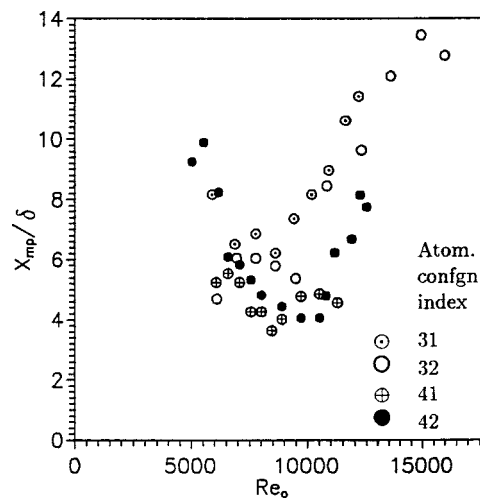


FIG. 12. Variation of X_{mp} at the merged flow rate condition against Re_o for different atomizer configurations.

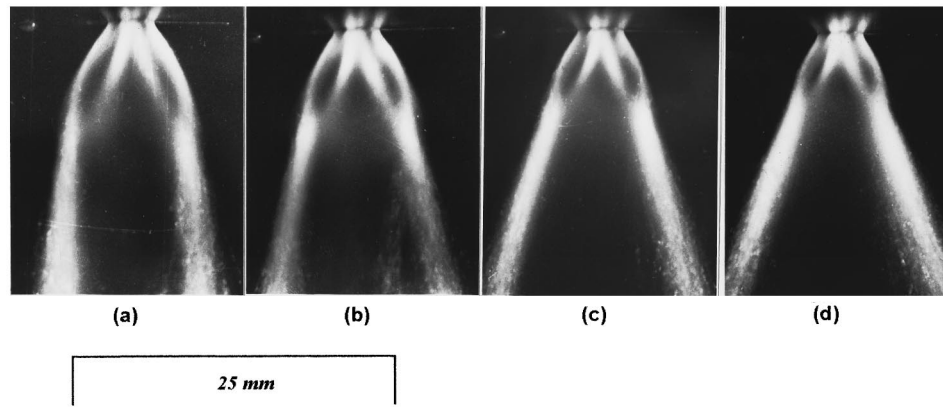


FIG. 13. Movement of the liquid sheet contact point with Re_i for a typical outer jet flow condition with $Re_o=5898$ (atomizer configuration index—31). (a) $Re_i=9465$, (b) $Re_i=10\,014$, (c) $Re_i=10\,536$, and (d) $Re_i=12\,454$.

data point corresponds to a distinct set of Re_o and Re_i . Going by a naive consideration of momenta, one would expect X_{mp} to increase with Re_o . However, it is observed from Fig. 12 that X_{mp} initially decreases, reaches a minimum, and thereafter increases with Re_o . The decreasing trend of X_{mp} with an increase in Re_o is attributed to the reduction in P_{int} inside the cavity, as discussed previously. At high Re_o , the drop in P_{int} inside the cavity through the entrainment process may not be sufficient to bend the outer liquid sheet possessing higher momentum. This reduces the net movement of the merging point toward the exit orifice during the transition process. As a consequence, the merging distance increases with increasing Re_o , as observed in Fig. 12.

4. Influence of flow conditions on X_{mp} after the merging process of liquid sheets

The photographs presented in Fig. 13 show the movement of the merging point toward the exit orifice plane as Re_i is increased, keeping Re_o constant at a rather low value. The quantitative variation of X_{mp} against Re_i is shown in Fig. 14(a) for different outer flow conditions. Here X_{mp} tends to decrease with increasing Re_i . But in case of a constant inner flow condition, X_{mp} tends to increase with increasing Re_o , as

seen in Fig. 14(b). These observations could be explained through the qualitative arguments given in the previous section.

Figure 15 shows the flow configuration of the jets during the separation process by reducing Re_i for a constant high Re_o . The first frame depicts the jet configuration at a merged condition, with a merging point close to the exit orifice plane. In addition to the shift in the contact point, an important feature in the second frame is that both the jets develop higher curvatures and, in fact, the inner jet becomes concave outward compared to the normal convex nature. The variation of X_{mp} during the separation process of liquid sheets is shown in Fig. 16 for different Re_o . It may be noticed that a small change in Re_i at the verge of the separation process makes the liquid sheet contact point move very rapidly away from the exit orifice plane followed by the separation of the liquid sheets.

5. Influence of geometric parameters on merging and separation processes of liquid sheets

Experiments have also been carried out for the low outer orifice gap ($d_o=2.7$ mm) configurations. In the case of high d_o atomizer configurations, just after the merging point, the combined jet disintegrates into irregular ligaments and then

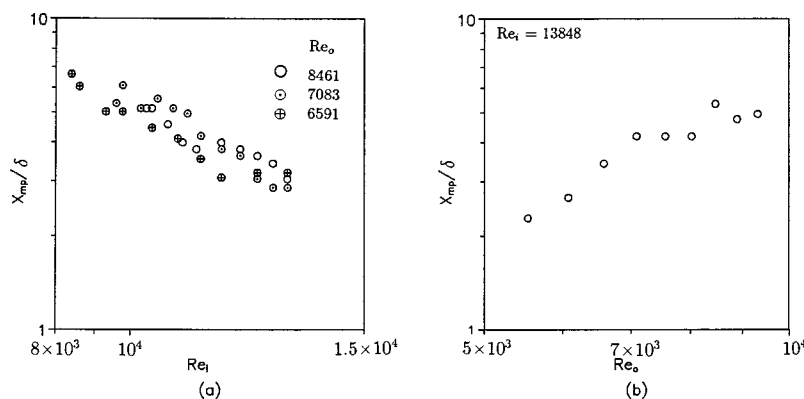


FIG. 14. The influence of flow conditions on X_{mp} (atomizer configuration index—42).

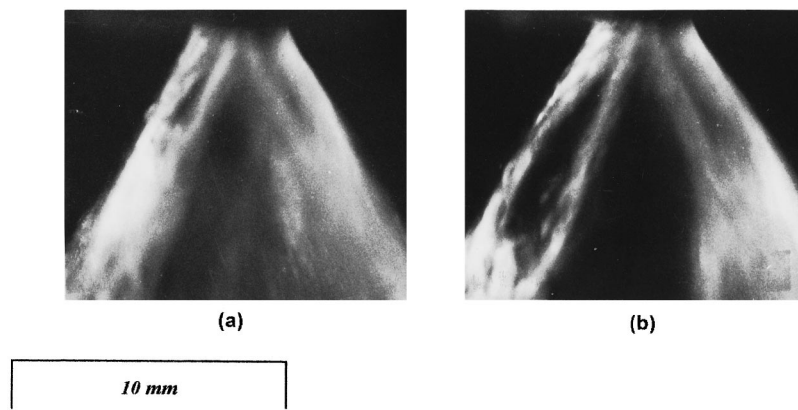


FIG. 15. Images of the movement of the liquid sheet contact point close to the separation process with $Re_o=11\,275$ (atomizer configuration index—42; the backlighting technique with a camera shutter speed of $1/1000$ s). (a) $Re_i=12\,168$ and (b) $Re_i=8990$.

into drops. For low d_o configurations as the present one, the liquid sheets merge very close to the exit orifice plane, and it is extremely difficult to discern the details of the cavity. The combined single liquid sheet formed after merging is seen in Fig. 17, which exhibits the characteristics of a single liquid sheet. The visualization of combined liquid sheets using a strobe light reveals that the resulting single liquid sheet has visibly different wave motion characteristics and more perforations over its surface.

As mentioned earlier, the variation in S is accomplished through the changes in swirler and exit orifice geometrical parameters. The effect of varying the swirl intensity of the jets on the merging flow rate condition in terms of the Re_i-Re_o plot is shown in Fig. 18. For a particular Re_o , the outer jet widens if the swirl intensity (represented by S_o) increases. A correspondingly higher cone angle of the inner jet and/or a greater reduction in the static pressure between the sheets is necessary for the merging to occur. This can be accomplished only by increasing Re_i , as observed in Fig. 18 for a particular Re_o . In contrast, the increase in the swirl intensity of the inner jet reduces the merging flow condition, as seen in Fig. 18, evidently due to the increase in the inner

jet spray cone angle. As may be observed, the differences due to changes in the geometrical parameters (through S_i or S_o) are larger at higher Re_o . A similar variation of the separation flow rate for the atomizer configurations mentioned in Fig. 18 have been plotted in Fig. 19. The separation state also exhibits the same kind of variation, as discussed above for the merging process.

The influence of the orifice recess defined by the offset distance between the inner and outer orifice exit, on the merging and separation behavior of the liquid sheets, has also been investigated. The corresponding variation of the merging flow rate condition is shown in Fig. 20 for different extents of recess. With an increase in recess, the merging flow rate reduces for a particular outer flow condition. Also, at high recess, it is observed from the figure that the merging flow rate is almost invariant with the outer jet flow rate. This trend has been visually observed by Seol *et al.*²¹ also, whose

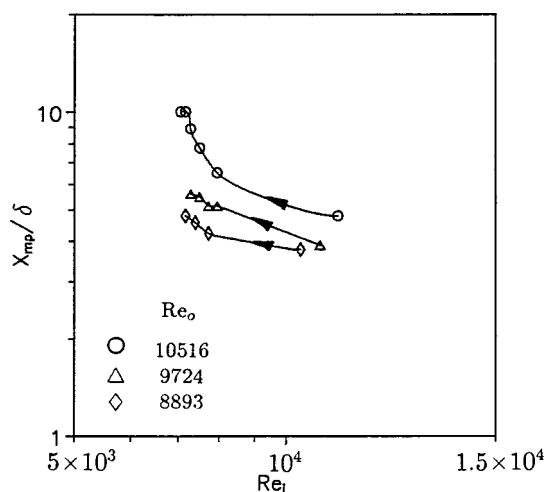


FIG. 16. The variation of X_{mp} during the separation process for different Re_o (atomizer configuration index—42).

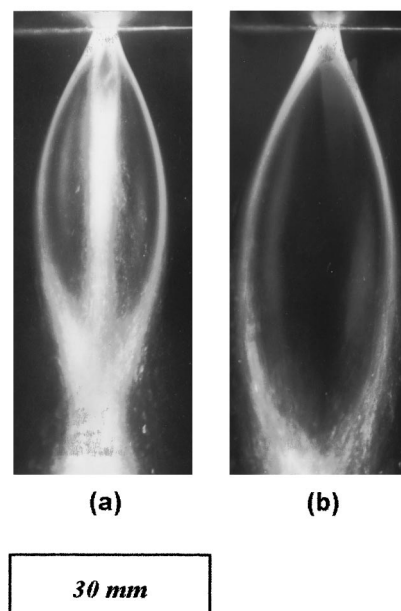


FIG. 17. Photographs of the separated and combined jet at low annular orifice gap, $Re_o=3460$ (atomizer configuration index—22). (a) $Re_i=6066$ and (b) $Re_i=6575$.

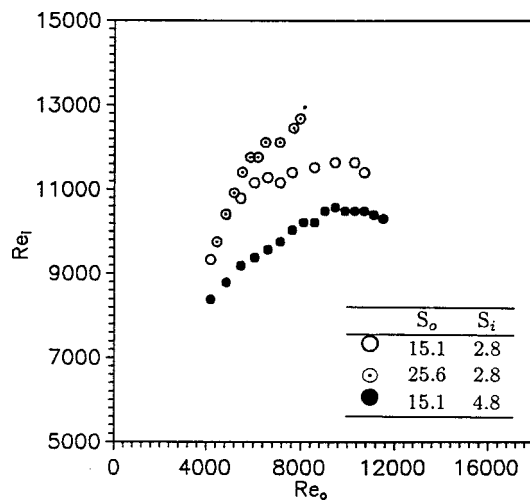


FIG. 18. The effect of jet swirl intensity on the merging flow condition.

experiments invariably involved the orifice recess. This also explains why the interaction phenomenon described earlier with coplanar configurations was not observed by Seol *et al.*²¹

6. The extent of the hysteresis regime

The hysteresis regime in which the liquid sheets are in either a merged or separated state, depending on how the flow conditions are attained, has been discussed earlier with reference to Fig. 10. The extent of the hysteresis regime $[2(Re_{i,mer} - Re_{i,sep}) / (Re_{i,mer} + Re_{i,sep})]$ is plotted in Fig. 21 against Re_o for different atomizer configurations. It is interesting to observe from the figure that the width of the hysteresis regime initially increases with Re_o , reaches a maximum, and thereafter decreases with Re_o . This behavior can be argued to be consistent with the cavity pressure and X_{mp} variations with Re_o shown in Fig. 12. The increasing trend of the hysteresis width with an increase in Re_o is attributed to the higher suction encountered in the cavity due to the reduction in P_{int} . This could be justified by the decreasing variation of X_{mp} with an increase in Re_o , as seen in Fig. 12.

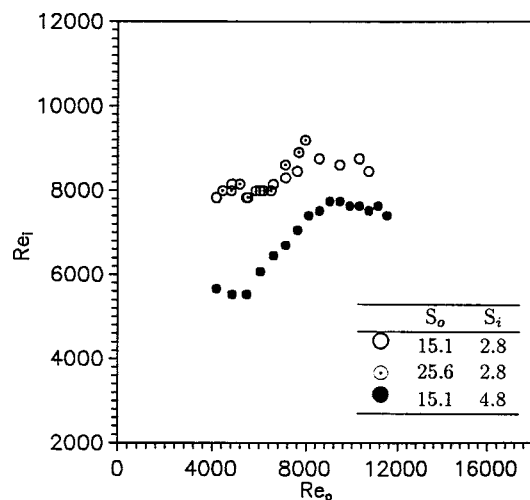


FIG. 19. The effect of jet swirl intensity on the separation flow condition.

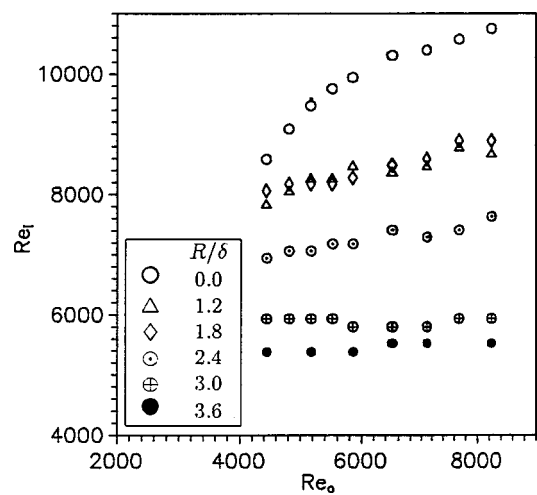


FIG. 20. The influence of the orifice recess on the merging flow condition (atomizer configuration index—22).

In a similar way, one can extend the argument of increasing variation of X_{mp} at higher Re_o (see Fig. 12) to the decreasing trend of the hysteresis width with an increase in Re_o . Therefore, it appears possible to reduce the extent of the hysteresis regime by increasing inertia (Re_o).

7. Influence of liquid surface tension on the merging and separation processes of liquid sheets

It was believed that the liquid surface tension, σ , pulls the sheets closer to the orifice exit such that the surface area is minimized.²² Experiments have been conducted by adding a surfactant to water to bring about a change in σ . The value of σ has been measured using stalagmometer and is found to decrease from 72.5×10^{-3} N/m for pure water to a range of 43.2×10^{-3} – 48.0×10^{-3} N/m with the addition of 0.07% of surfactant. The influence of σ on the merging and separation flow condition is shown in Fig. 22. It is seen that the reduction in σ reduces the merging and separation flow conditions only marginally. In a swirling liquid jet, the centrifugal force is balanced by the pressure differential, $(P_e - P_i)$, plus sur-

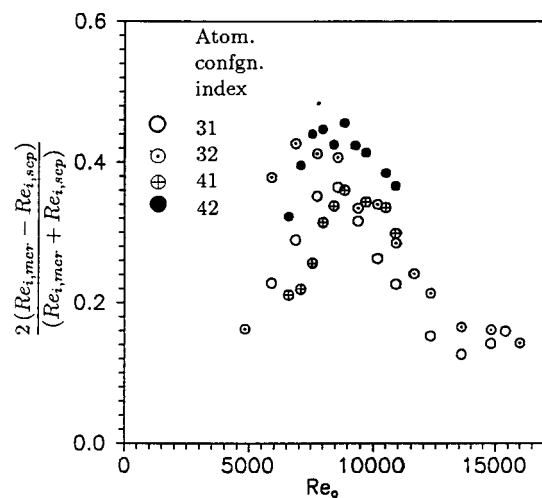


FIG. 21. The extent of the hysteresis regime for different atomizer configurations.

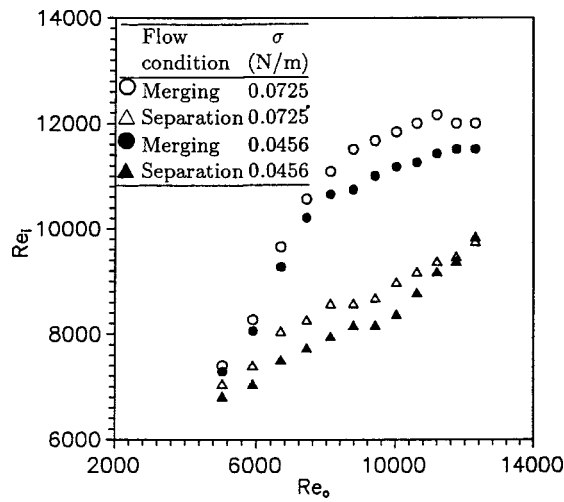


FIG. 22. The influence of liquid surface tension on the merging and separation flow conditions (atomizer configuration index—32).

face tension, which is acting radially inward. For a constant centrifugal force, the reduction in surface tension leads to an increase in $(P_e - P_i)$. As a consequence, the liquid sheets merge at slightly reduced flow conditions, as observed in Fig. 22, due to a minor reduction in P_{int} . However, a further justification of this qualitative argument requires more information on the static pressure variation between the liquid sheets. The influence of σ on X_{mp} variation is shown in Fig. 23. It is inferred that the reduction in σ slightly moves the merging point toward the orifice plane.

E. Influence of the merging and separation behavior of liquid sheets on the spray characteristics

As was seen in Fig. 11, the combined liquid jet could be in different states in regime III depending on how that regime is approached. This can be expected to result in drastic variation in spray parameters like spray width, drop sizes that could have a significant influence on the combustion efficiency of any engine employing such atomizers.

1. Spray width variation

The following discussion on spray width may be viewed as that on the effective cone angle of the outer jet. The variation of SW_{10} , which is the typical measurement location to

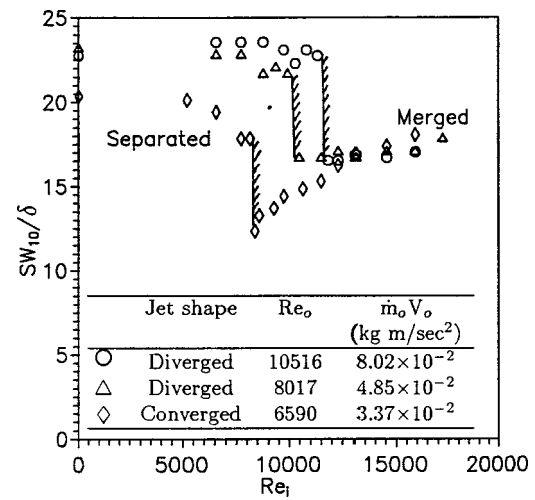


FIG. 24. The plot showing the sudden drop in spray width during the merging process at 10 mm distance from the exit of the orifice (atomizer configuration index—42).

determine the spray cone angle²⁶ with Re_i is plotted in Fig. 24 for different outer jet flow conditions. For a diverging outer jet, increase in Re_i reduce the spray width, with a jump at the merging condition. However, for a converging outer jet (low Re_o), there is a gradual reduction seen in the separated flow regime before the sudden drop. After merging, the spray width increases and gradually attains the same state as the first case. The gradual variation of spray width at low Re_o is primarily attributed to the lower momentum of the liquid sheets, whence the reduction in P_{int} is sufficient to produce the curvature in the liquid sheet. It is also observed from the figure that the influence of Re_o on SW_{10} is negligible at high Re_i after merging. This shows that the behavior of the combined jet is primarily controlled by the inner jet flow conditions. This could be attributed to the higher momentum of the inner jet in comparison with the outer jet. For the present case, the exit momentum of the inner liquid sheet is much higher (at $Re_i=11\,515$, $\dot{m}_i V_i=5.41 \times 10^{-2}$ kg m/s²; and at $Re_i=16\,589$, $\dot{m}_i V_i=10.62 \times 10^{-2}$ kg m/s²) compared to that of the outer jet momentum (see Fig. 24). The gradual increase in the spray width after merging at low Re_o could also be explained by the above arguments. Figure 25 shows the quantitative variation of spray width (SW_{10}) plotted around

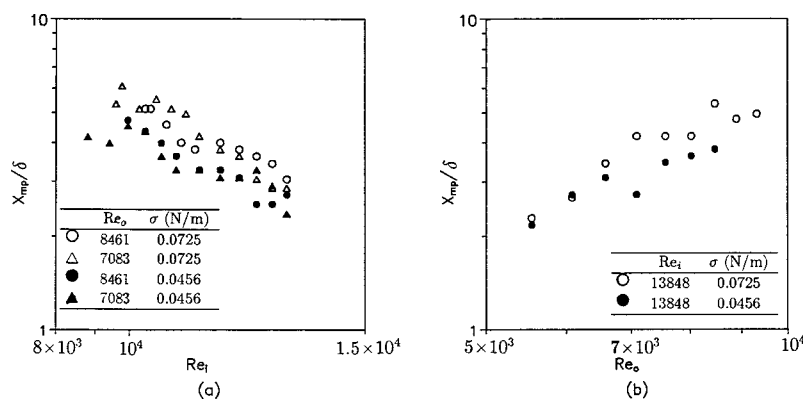


FIG. 23. The effect of liquid surface tension on X_{mp} variation with flow conditions (atomizer configuration index—42).

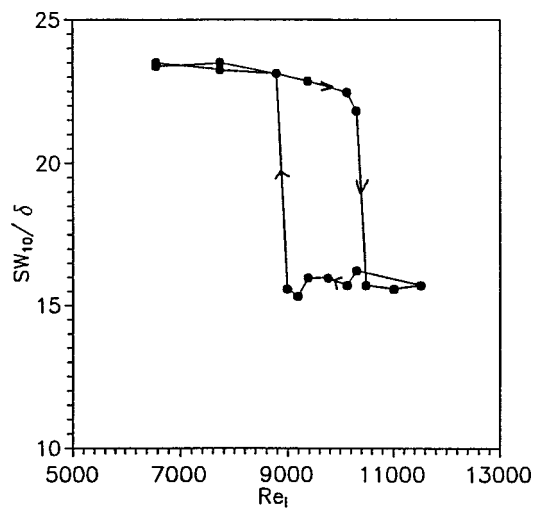


FIG. 25. The influence of hysteresis in the process of the merging and separation of the liquid sheets on the spray width of the combined jet with $Re_o = 9314$ (atomizer configuration index—42).

the hysteresis loop against Re_i for a typical outer flow condition. This clearly demonstrates the existence of different values of spray width, depending on how the flow condition is approached.

2. Drop size variation

Measurements have been carried out to get the SMD variations of the spray in regime III around the hysteresis loop using a Malvern particle sizer. The variation of SMD in regime III with changes in the inner jet flow condition is shown in Fig. 26 for two different atomizer configurations with constant Re_o . The plot reveals that there is a significant increase in SMD value due to merging of the liquid sheets. In the merged state, the mixing of two sprays would lead to an increase in droplet sizes of the combined spray along the branch "R" in Fig. 26. This is perhaps due to an effectively thicker sheet before disintegration and also droplet coalescence and collisions in the resultant dense spray zone. It is

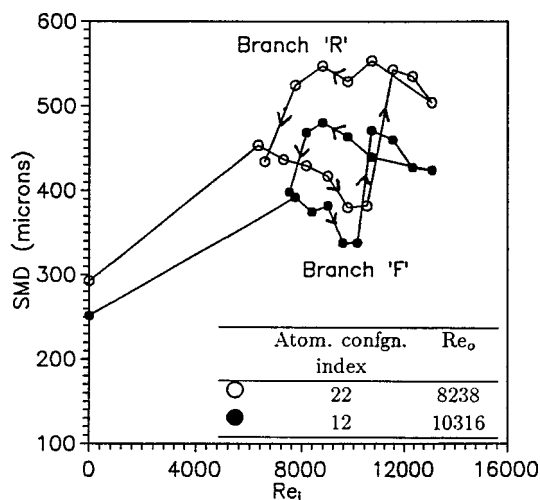


FIG. 26. The influence of hysteresis in the process of the merging and separation of the liquid sheets on SMD of the global spray.

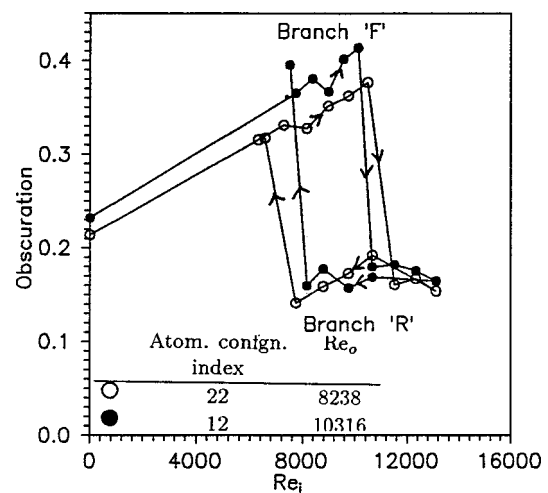


FIG. 27. The influence of the hysteresis in the process of the merging and separation of the liquid sheets on the obscuration level of the global spray.

observed that the presence of the inner jet in the core of the outer jet in the separated mode (branch "F") causes little change in the distribution or SMD from that corresponding to the total absence of the inner jet. The variation of the obscuration level, which is a measure of the liquid concentration in the sampling region of the laser beam path, as plotted in Fig. 27, shows a sudden drop as merging occurs. An equally sudden rise occurs as the jets separate at a lower Re_i . At a particular Re_i in the hysteresis region, the total liquid flow rate is constant, though obscuration levels are distinct. The reduction in the obscuration level during the merging phenomenon may be explained by the increase in the droplet diameter and the resultant decrease in the total surface area of the droplets in the spray.

IV. CONCLUSIONS

The interaction of thin conical liquid sheets formed from liquid-liquid coaxial swirl atomizers are studied using water and water with added surfactant as the experimental fluids. The conical sheets are visualized by illuminating a section of the jets by a collimated sheet of a light and backlighting technique. A fluorescent dye is mixed with the experimental fluid to increase the contrast of the spray contour. The major conclusions of the study are the following.

- (1) The presence of an inner jet in the core of the outer jet alters its spray parameters like the spray width, the disintegration distance of the liquid sheet, the tulip length, etc., even before the merging of liquid sheets occurs. The pressure variation between the liquid sheets is inferred to be primarily responsible for this phenomenon.
- (2) The transition process from a separated to a merged state is very rapid, depending on the flow conditions. The separation of liquid sheets also exhibits similar behavior.
- (3) The process of merging and separation of the two conical liquid sheets exhibits a hysteresis. The extent of hysteresis could be controlled by the inertia and/or swirl intensity of the liquid sheets.
- (4) During the merging process, the liquid sheets come in contact at a point farther downstream of the exit orifice

plane. Further movement of this merging point is controlled by the flow conditions. The pressure in the cavity between the liquid sheets can explain many of the observed trends.

(5) Geometrical parameters like exit diameters of the orifices, swirler passage dimensions, and the orifice recess affect the merging and separation process of the liquid sheets. The merging flow condition is slightly reduced with the decrease in liquid surface tension.

(6) The combined spray exhibits a drastic change in its spray parameters like spray width, drop size distribution, disintegration behavior, etc., during the transition process from the separated to the merged state, and vice versa. Also, the spray parameters of the combined spray are influenced by the hysteresis phenomenon, which, in turn, can strongly influence the combustion efficiencies.

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